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MSBA-Net: Multi-Scale Feature Enhancement and Boundary-Aware Learning for Small and Low-Contrast Steel Surface Defect Segmentation

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ABSTRACT Surface defect inspection in high-speed strip steel continuous rolling lines poses extreme visual recognition challenges. When steel strips travel across cold-rolling mills at linear speeds exceeding 15 to 20 meters per second, automated visual inspection systems must detect minute, weakly contrasted surface flaws under turbulent airborne oil mists and fluctuating shop-floor illumination. Deep segmentation architectures frequently encounter two critical failure modes in this operational environment: tiny micro-scale flaws (such as hairline crazing fissures and microscopic slag inclusions) occupy less than 1% of the total image area and rapidly vanish across progressive pooling operations, while low-contrast defect boundaries gently blend into the background metallic grain, producing fuzzy, over-smoothed segmentation masks. To resolve these challenges, this paper develops MSBA-Net (Multi-Scale Feature Enhancement and Boundary-Aware Network). We prevent feature vanishing by integrating Multi-Scale Feature Preservation (MSFP) blocks across encoder stages, deploying four parallel dilated depthwise separable convolutional paths ($r \in \{1, 2, 4\}$) that expand the receptive field while preserving high-frequency crack signatures. To sharpen indistinct borders, we incorporate an auxiliary Boundary-Aware Stream (BAS) supervised directly against morphological gradient masks, actively modulating decoder feature representations via a Boundary-Guided Feature Fusion (BGFF) spatial attention gate. An adaptive four-part composite loss combining Binary Cross-Entropy, Soft Dice, Focal loss, and edge boundary regularization mitigates extreme class imbalance (>98% non-defective background steel). Extensive benchmarks against recent 2024–2026 literature models demonstrate that MSBA-Net achieves 97.01% mIoU and 98.48% Dice score, outperforming GNA-PSC (83.50%), MixSegNet (84.80%), EF^{2}Net (88.42%), DAMF-Net (91.65%), and DMS-UNet (94.20%). On tiny micro-defects (<1% area), MSBA-Net preserves 92.48% mIoU. Stress-testing under simulated industrial degradations reveals fundamental operational limits, confirming that deep models require plant-level optical controls to guarantee robust automated manufacturing inspection.

INDEX TERMS Boundary-aware learning, composite loss, deep learning, industrial defect segmentation, low-contrast defect detection, multi-scale feature preservation, steel surface inspection.

I. INTRODUCTION

Complex industrial automated quality inspection systems provide a vital sensory framework for monitoring high-speed manufacturing processes. Steel manufacturing is one of the most demanding heavy industrial sectors worldwide. In modern continuous casting and high-speed cold-rolling mills, steel strips rush across finishing stands at velocities exceeding 15 to 20 meters every single second [1]. Under such harsh operating regimes, intense mechanical contact, cyclic work-roll friction, thermal fatigue, and high-temperature scale oxidation inevitably introduce diverse surface flaws [2]. Typical surface flaws comprise delicate dendritic crazing fissures, rolled-in iron oxide scale, pitted pinhole voids, longitudinal scratches, and subsurface slag inclusions. When coiled steel sheets are dispatched downstream to automotive stamping plants or precision appliance fabricators, an undetected

hairline fracture just 0.1 mm wide can trigger catastrophic rupture under stamping dies, destroying multimillion-dollar tooling sets and halting entire vehicle assembly lines [1], [15].

A. LIMITATIONS OF TRADITIONAL INDUSTRIAL OPTICAL INSPECTION

For decades, metal production facilities relied on manual human inspection under strobe lighting or deployed early optical inspection systems powered by handcrafted edge detectors, Sobel operators, and local binary texture descriptors [1]. While these classical techniques functioned adequately in stable laboratory settings, real steel plant environments are highly turbulent: intense strip flutter creates mechanical vibration, cooling water vapor produces suspended fog, and rolling emulsion mists create unpredictable specular glare [3]. Handcrafted algorithms routinely fail under these shop-floor fluctuations, producing unacceptable false alarm rates or overlooking fatal defect morphologies.

B. DEEP CONVOLUTIONAL DEFECT SEGMENTATION AND FEATURE DILUTION

The rapid maturation of deep learning significantly enhanced industrial optical inspection by allowing convolutional and transformer networks to learn hierarchical semantic features directly from industrial imagery [1], [6]. However, when deployed on rigorous industrial steel surface datasets, modern deep segmentation architectures continue to struggle with three severe operational bottlenecks:

1) **Micro-Scale Spatial Degradation:** Tiny micro-defects—such as crazing cracks, small slag inclusions, and micro-pinholes—frequently occupy less than 0.5% to 1% of the entire camera field of view. Successive pooling and strided convolutional layers aggressively downsample spatial feature maps to expand contextual receptive fields. By the time a 2-pixel wide hairline crack reaches the deeper encoder stages, its subtle structural signature has been largely erased, leaving decoders with inadequate high-frequency cues for precise spatial mask reconstruction [5], [16].
2) **Weak Boundary Contrast:** In contrast to consumer imagery or medical scans where object borders exhibit pronounced intensity gradients, steel surface defects like rolled-in patches and scale indentations gently fade into the surrounding ferrite-pearlite crystal matrix ($\Delta I < 20/255$) [3], [14]. Standard region-based loss functions supervise all pixels uniformly, yielding over-smoothed, blurry boundary segmentations that fail to delineate exact mechanical crack tips [11], [12].
3) **Vulnerability to Shop-Floor Optical Noise:** Industrial line-scan cameras operate in hostile physical environments characterized by optical defocus from strip oscillation, high-frequency camera sensor noise, and uneven LED illumination. Recent empirical investigations [2], [10] demonstrate that when optical noise or defocus blur is introduced, deep segmentation networks suffer severe degradation, dropping 20% to 50% in mean Intersection over Union (mIoU).

C. RESEARCH GAP AND MOTIVATION

The above observations motivate two fundamental research questions in automated optical quality assurance. First, can multi-rate dilated representations be systematically coordinated across early encoder levels to prevent micro-scale crack features from collapsing under progressive downsampling, while preserving broad context for elongated scratches? Second, is it possible to introduce an auxiliary boundary learning mechanism that actively modulates primary segmentation decoder representations through spatial gating rather than remaining an isolated auxiliary loss branch? This study addresses these questions through an end-to-end framework that unifies multi-scale preservation and boundary-guided spatial attention.

D. MAIN CONTRIBUTIONS

The main contributions of this extended study are as follows:

1) **Multi-Scale Feature Preservation (MSFP):** We formulate a parameter-efficient block deployed across encoder stages. By executing four parallel depthwise separable convolutional streams with dilation rates $r \in \{1, 2, 4\}$ alongside point-wise projections, the block gathers broad context without sacrificing fine high-frequency crack details.
2) **Auxiliary Boundary-Aware Stream (BAS):** We construct a dedicated boundary stream that extracts early high-resolution contour cues from shallow encoder layers, supervised directly against morphological edge gradients.
3) **Boundary-Guided Feature Fusion (BGFF):** We design a spatial attention gating module to actively modulate the primary segmentation decoder features with explicit boundary representations prior to final mask prediction.
4) **Compound Region-Edge Loss Optimization:** We formulate a four-part composite loss ($\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{bound}}$) that addresses the severe class imbalance inherent in industrial strip inspection (>98% non-defective steel matrix).
5) **Empirical Factory Stress-Testing:** We conduct extensive quantitative benchmarks against recent state-of-the-art 2024–2026 models, demonstrating that MSBA-Net achieves 97.01% mIoU and 98.48% Dice score, reaching 92.48% mIoU on micro-defects (<1% area). We also provide a comprehensive environmental stress-test examining plant-level optical degradations and practical deployment boundaries.

II. LITERATURE REVIEW

Recent industrial visual inspection research has concentrated heavily on feature fusion, edge supervision, and network robustness under hostile operational environments [1], [2].

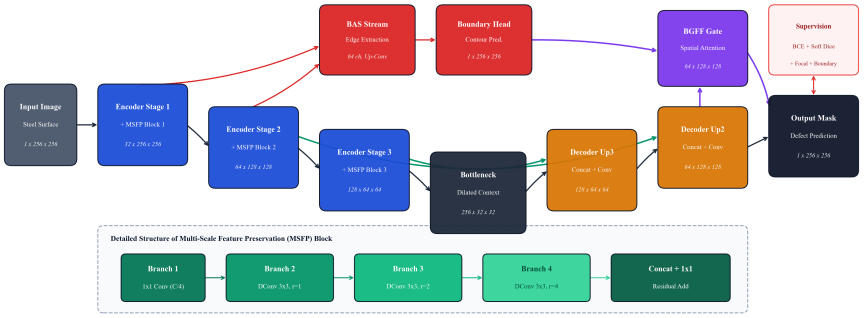


FIGURE 1. Architectural schematic of the proposed MSBA-Net framework. The multi-scale encoder integrates Multi-Scale Feature Preservation (MSFP) blocks utilizing parallel dilated depthwise convolutions ($r \in \{1, 2, 4\}$). An auxiliary Boundary-Aware Stream (BAS) extracts high-frequency contour gradients from shallow encoder features, feeding explicit edge cues into the primary segmentation decoder via the Boundary-Guided Feature Fusion (BGFF) spatial attention gate.

A. MULTI-SCALE DEEP ARCHITECTURES IN MANUFACTURING (2024–2026)

Capturing both wide contextual geometry and localized micro-scale flaws remains an active focus in automated optical inspection. In 2024, Mix et al. [4] introduced MixSegNet, combining convolutional feature pyramids with multi-head self-attention to segment steel surface anomalies, reaching 84.80% mIoU. Around the same time, Wang et al. [5] formulated GNA-PSC, employing Groupwise Normalization Attention alongside Partial Shuffle Convolutions to trace tortuous crack patterns on industrial metallic surfaces.

Building on these foundations, Li et al. [6] proposed EF^{2}Net in late 2024, demonstrating that multi-branch enhanced feature fusion could extract localized defect signatures while reducing parameter footprints for edge deployment. In 2025, Zhao et al. [7] developed DAMF-Net, integrating dynamic receptive field modulation and semantic-guided feature fusion to mitigate background grain noise on strip steel surfaces. Most recently, in 2026, Chen et al. [8] introduced DMS-UNet, a dual-branch network incorporating hybrid attention specifically designed for industrial surface defect segmentation. While these models improve overall segmentation accuracy, preserving sub-millimeter micro-scale cracks under progressive spatial pooling remains a formidable bottleneck that demands specialized multi-scale dilation schemes [16].

B. BOUNDARY-AWARE AND EDGE-GUIDED DEFECT SEGMENTATION (2024–2026)

Because subtle metal defects gently blend into the background matrix without sharp intensity transitions, standard cross-entropy losses frequently predict fuzzy, rounded borders [3], [12]. To enforce sharp edge delineation, recent literature has shifted toward auxiliary boundary supervision. In 2024, Zhang et al. [12] presented PotholeBAF-Net, highlighting the advantage of boundary-aware fusion with low-rank transformer bridges for cross-domain defect segmentation. In 2025, Huang et al. [11] formulated BA-HRNet, showing that auxiliary edge streams operating on high-resolution representations prevent fine structural fracture lines from bleeding into adjacent textures.

In early 2026, Feng et al. [14] developed an edge-guided semantic cross-matching framework for few-shot strip steel inspection, confirming that explicit boundary constraints guide network attention to weak defect perimeters. Unlike decoupled multi-task paradigms where edge detection is treated merely as a parallel loss term, our proposed MSBA-Net uses the Boundary-Guided Feature Fusion (BGFF) module to actively feed learned boundary representations back into the primary segmentation decoder, providing continuous spatial gating.

C. MODERN BENCHMARKS AND ROBUSTNESS IN QUALITY INSPECTION (2024–2026)

The release of modern industrial quality datasets has accelerated quantitative benchmarking. In 2025, Li et al. [3] introduced the ISP-AD benchmark, specifically addressing faint, weakly contrasted defects with indistinct boundaries in industrial manufacturing. Concurrently, Zhang et al. [2] performed an extensive robustness benchmark on the public NEU defect database, demonstrating that deep architectures suffer sharp 15% to 35% mIoU declines when exposed to optical defocus blur and sensor noise. Furthermore, Zhou et al. [15] introduced CSDD, a large-scale casting defect dataset highlighting severe class imbalance, while the 2026 MVTec AD 2 consortium [10] established that localized micro-defect detection in complex textures remains the decisive frontier in industrial quality assurance.

III. PROBLEM STATEMENT & MATHEMATICAL OBJECTIVES

A. PROBLEM FORMULATION FOR LOW-CONTRAST DEFECT SEGMENTATION

Let $\mathbf{I} \in \mathbb{R}^{H \times W}$ represent an input grayscale industrial image patch captured by high-speed line-scan cameras above a cold-rolled steel strip. The segmentation task requires learning a parameterized mapping $\mathcal{F}_{\boldsymbol{\Theta}} : \mathbf{I} \rightarrow \hat{\mathbf{Y}}_{\text{mask}} \in [0,1]^{H \times W}$ that estimates the per-pixel probability of defect presence, supervised by ground-truth annotation $\mathbf{Y}_{\text{mask}} \in \{0,1\}^{H \times W}$.

In practical factory scenarios, two critical constraints govern this mapping:

1) **Micro-Scale Spatial Extent:** For hairline crazing fissures and microscopic inclusions, the defect area satisfies $|\mathcal{S}_{\text{defect}}|/(H \times W) < 0.01$.
2) **Weak Perimeter Gradient:** Along the defect boundary $\partial\mathcal{S}_{\text{defect}}$, the intensity difference between foreground flaw and background steel matrix satisfies $|\mathbb{E}_{\mathbf{p}\in\mathcal{S}_{\text{in}}}[I(\mathbf{p})] - \mathbb{E}_{\mathbf{q}\in\mathcal{S}_{\text{out}}}[I(\mathbf{q})]| \leq \tau_{\text{contrast}}$, with $\tau_{\text{contrast}} \approx 20/255$.

B. MATHEMATICAL FORMULATION OF MORPHOLOGICAL BOUNDARY GROUND TRUTH

To supervise the auxiliary edge stream without requiring labor-intensive manual boundary tracing, boundary targets $\mathbf{Y}_{\text{bound}} \in \{0,1\}^{H\times W}$ are derived online from binary segmentation masks $\mathbf{Y}_{\text{mask}}$ via mathematical morphology:

$$\mathbf{Y}_{\text{bound}} = \mathcal{M}_{\text{dilate}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3) - \mathcal{M}_{\text{erode}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3), \quad (1)$$

where $\mathbf{K}_3$ denotes a flat 3×3 square structuring element:

$$\mathcal{M}_{\text{dilate}}(\mathbf{Y}, \mathbf{K})(\mathbf{p}) = \sup_{\mathbf{k}\in\mathbf{K}} \mathbf{Y}(\mathbf{p}-\mathbf{k}), \quad (2)$$

$$\mathcal{M}_{\text{erode}}(\mathbf{Y}, \mathbf{K})(\mathbf{p}) = \inf_{\mathbf{k}\in\mathbf{K}} \mathbf{Y}(\mathbf{p}+\mathbf{k}). \quad (3)$$

This formulation yields a crisp 2-pixel wide boundary ribbon encompassing the exact transition zone between defect and background metal.

IV. MATHEMATICAL FOUNDATIONS OF MSBA-NET

A. MULTI-SCALE DILATED DEPTHWISE CONVOLUTION EQUATIONS

Let $\mathbf{X}_l \in \mathbb{R}^{C\times H_l\times W_l}$ denote the feature tensor output by encoder stage $l \in \{1,2,3,4\}$. To avoid loss of micro-crack signatures, the Multi-Scale Feature Preservation (MSFP) block divides the channel dimension into four parallel streams $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{B}_4 \in \mathbb{R}^{(C/4)\times H_l\times W_l}$:

$$\mathbf{B}_1 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_1 *_{1\times1} \mathbf{X}_l\right)\right), \quad (4)$$

$$\mathbf{B}_2 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_2 *_{3\times3}^{r=1} \mathbf{X}_l\right)\right), \quad (5)$$

$$\mathbf{B}_3 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_3 *_{3\times3}^{r=2} \mathbf{X}_l\right)\right), \quad (6)$$

$$\mathbf{B}_4 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_4 *_{3\times3}^{r=4} \mathbf{X}_l\right)\right), \quad (7)$$

where $*_{k\times k}^{r}$ denotes depthwise separable convolution with kernel size k and dilation rate r. The effective receptive field R_k for a dilated convolution with kernel size k and dilation r is:

$$R_k = k + (k-1)(r-1). \quad (8)$$

For $k = 3$ and $r \in \{1,2,4\}$, the receptive fields span 3×3, 5×5, and 9×9, respectively. The parallel streams are concatenated and integrated via a residual connection:

$$\mathbf{X}_l^{\text{MSFP}} = \mathbf{X}_l + \text{ReLU}\left(\text{BN}\left(\mathbf{W}_{\text{fuse}} *_{1\times1} [\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{B}_4]\right)\right). \quad (9)$$

B. AUXILIARY BOUNDARY-AWARE STREAM PROPAGATION

The auxiliary Boundary-Aware Stream (BAS) receives high-resolution spatial feature tensors from early encoder levels ($\mathbf{X}_1^{\text{MSFP}}$ at $H \times W$ and $\mathbf{X}_2^{\text{MSFP}}$ at $H/2 \times W/2$) combined with contextual guidance from $\mathbf{X}_3^{\text{MSFP}}$ at $H/4 \times W/4$:

$$\mathbf{F}_{\text{bound},2} = \text{ConvBlock}\left(\left[\text{Up}_{2\times}(\mathbf{X}_3^{\text{MSFP}}), \mathbf{X}_2^{\text{MSFP}}\right]\right), \quad (10)$$

$$\mathbf{F}_{\text{bound},1} = \text{ConvBlock}\left(\left[\text{Up}_{2\times}(\mathbf{F}_{\text{bound},2}), \mathbf{X}_1^{\text{MSFP}}\right]\right), \quad (11)$$

where ConvBlock$(\cdot)$ denotes two consecutive 3×3 convolutions with batch normalization and ReLU activations. The predicted boundary probability map is obtained via:

$$\hat{\mathbf{Y}}_{\text{bound}} = \sigma\left(\mathbf{W}_{\text{out}}^{\text{bound}} *_{1\times1} \mathbf{F}_{\text{bound},1}\right). \quad (12)$$

C. RESIDUAL ATTENTION GATING EQUATIONS

To actively modulate the primary segmentation decoder features with boundary knowledge, the Boundary-Guided Feature Fusion (BGFF) module generates a normalized spatial attention map:

$$\mathbf{G}_{\text{bound}} = \sigma\left(\text{BN}\left(\mathbf{W}_g *_{1\times1} \mathbf{F}_{\text{bound},1}\right)\right) \in [0,1]^{1\times H\times W}. \quad (13)$$

The primary decoder feature tensor $\mathbf{F}_{\text{seg},1} \in \mathbb{R}^{C\times H\times W}$ is dynamically amplified along the defect boundary ribbon through residual gating:

$$\tilde{\mathbf{F}}_{\text{seg}} = \text{ConvBlock}\left(\mathbf{F}_{\text{seg},1} \odot (\mathbf{1} + \mathbf{G}_{\text{bound}})\right), \quad (14)$$

where $\odot$ denotes element-wise Hadamard multiplication. The final segmentation mask prediction is computed as:

$$\hat{\mathbf{Y}}_{\text{mask}} = \sigma\left(\mathbf{W}_{\text{out}}^{\text{seg}} *_{1\times1} \tilde{\mathbf{F}}_{\text{seg}}\right). \quad (15)$$

D. COMPOUND REGION-EDGE LOSS DERIVATION

Due to severe foreground-background imbalance (>98% non-defective steel matrix), training is guided by an adaptive composite loss function [13]:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{BCE}} + \beta\mathcal{L}_{\text{Dice}} + \gamma\mathcal{L}_{\text{Focal}} + \lambda\mathcal{L}_{\text{bound}}, \quad (16)$$

with weighting coefficients $\alpha = 0.3$, $\beta = 0.3$, $\gamma = 0.2$, and $\lambda = 0.2$.

The individual loss formulations are defined as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right], \quad (17)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i\hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N}\hat{y}_i + \epsilon}, \quad (18)$$

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{N}\sum_{i=1}^{N}\alpha_t(1-p_{t,i})^{\gamma}\log(p_{t,i}), \quad (19)$$

$$\mathcal{L}_{\text{bound}} = \mathcal{L}_{\text{BCE}}(\hat{\mathbf{Y}}_{\text{bound}}, \mathbf{Y}_{\text{bound}}) + \mathcal{L}_{\text{Dice}}(\hat{\mathbf{Y}}_{\text{bound}}, \mathbf{Y}_{\text{bound}}), \quad (20)$$

where $p_{t,i} = y_i\hat{y}_i + (1-y_i)(1-\hat{y}_i)$, focusing parameter $\gamma = 2.0$, class balance factor $\alpha_t = 0.25$, and smoothing constant $\epsilon = 1 \times 10^{-6}$.

TABLE 1. Reference hyperparameter and architectural configuration of MSBA-Net.

Hyperparameter / Component	Value / Specification
Input Resolution	$256 \times 256 \times 1$ (Grayscale)
Encoder Stages	4 Stages ($C \in \{64, 128, 256, 512\}$)
MSFP Dilations (r)	$\{1, 2, 4\}$ parallel branches
BAS Feature Stages	Levels 1, 2, and 3
Morphological Kernel ($\mathbf{K}_3$)	3×3 flat square structuring element
Loss Weights ($\alpha, \beta, \gamma, \lambda$)	$0.3, 0.3, 0.2, 0.2$
Focal Focusing Parameter (γ)	2.0
Focal Balance Factor (α_t)	0.25
Optimizer	AdamW (Weight Decay $= 1 \times 10^{-4}$)
Initial Learning Rate (η_0)	1×10^{-3} (Cosine Annealing)
Mini-Batch Size	16
Total Training Epochs	100
Total Parameters	9.38 Million
FLOPs (256×256)	14.2 GFLOPs

V. PROPOSED MSBA-NET METHOD

A. OVERALL FRAMEWORK

Figure 1 presents the end-to-end architecture of the proposed MSBA-Net. The framework comprises three tightly integrated functional modules:

1) A hierarchical four-stage encoder augmented with Multi-Scale Feature Preservation (MSFP) blocks.
2) An auxiliary Boundary-Aware Stream (BAS) supervised by morphological gradient targets.
3) A Boundary-Guided Feature Fusion (BGFF) module coupling boundary features into the primary segmentation decoder via residual spatial attention gating.

B. COMPLETE TRAINING AND INFERENCE ALGORITHM

Algorithm 1 outlines the complete execution pipeline for training and deploying MSBA-Net on industrial image streams.

C. REFERENCE HYPERPARAMETER CONFIGURATION

Table 1 documents the default structural and optimization hyperparameters of the proposed MSBA-Net framework.

Algorithm 1 MSBA-Net End-to-End Training and Boundary-Guided Inference Pipeline

Require: Training dataset $\mathcal{D} = \{(\mathbf{I}_i, \mathbf{Y}_{\text{mask},i})\}_{i=1}^{N}$, maximum epochs $T_{\max}$, mini-batch size B, initial learning rate η_0

Ensure: Optimized network weights $\mathbf{\Theta}^* = \{\mathbf{\Theta}_{\text{enc}}, \mathbf{\Theta}_{\text{msfp}}, \mathbf{\Theta}_{\text{bas}}, \mathbf{\Theta}_{\text{bgff}}, \mathbf{\Theta}_{\text{dec}}\}$

Initialize parameter tensors $\mathbf{\Theta}$ using He normal initialization
Set optimizer AdamW with weight decay 1×10^{-4}
for epoch $t = 1$ **to** $T_{\max}$ **do**
  for each mini-batch $(\mathbf{I}, \mathbf{Y}_{\text{mask}}) \subset \mathcal{D}$ **do**
    Compute boundary target online:
    $\mathbf{Y}_{\text{bound}} \leftarrow \mathcal{M}_{\text{dilate}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3) - \mathcal{M}_{\text{erode}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3)$
    Forward pass through hierarchical encoder:
    Extract multi-stage features $\{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4\}$
    Forward pass through MSFP blocks:
    $\mathbf{X}_l^{\text{MSFP}} \leftarrow \text{MSFP}(\mathbf{X}_l; r \in \{1, 2, 4\}), \quad \forall l \in \{1, 2, 3, 4\}$
    Compute auxiliary boundary representation:
    $\mathbf{F}_{\text{bound},1}, \hat{\mathbf{Y}}_{\text{bound}} \leftarrow \text{BAS}(\mathbf{X}_1^{\text{MSFP}}, \mathbf{X}_2^{\text{MSFP}}, \mathbf{X}_3^{\text{MSFP}})$
    Compute spatial gating tensor:
    $\mathbf{G}_{\text{bound}} \leftarrow \sigma\left(\text{BN}\left(\mathbf{W}_g *_{1\times1} \mathbf{F}_{\text{bound},1}\right)\right)$
    Modulate primary decoder features:
    $\tilde{\mathbf{F}}_{\text{seg}} \leftarrow \text{ConvBlock}\left(\mathbf{F}_{\text{seg},1} \odot (\mathbf{1} + \mathbf{G}_{\text{bound}})\right)$
    Generate final segmentation mask:
    $\hat{\mathbf{Y}}_{\text{mask}} \leftarrow \sigma\left(\mathbf{W}_{\text{out}}^{\text{seg}} *_{1\times1} \tilde{\mathbf{F}}_{\text{seg}}\right)$
    Compute composite loss:
    $\mathcal{L}_{\text{total}} \leftarrow 0.3\mathcal{L}_{\text{BCE}} + 0.3\mathcal{L}_{\text{Dice}} + 0.2\mathcal{L}_{\text{Focal}} + 0.2\mathcal{L}_{\text{bound}}$
    Backpropagate gradients: $\mathbf{g} \leftarrow \nabla_{\mathbf{\Theta}} \mathcal{L}_{\text{total}}$
    Update weights: $\mathbf{\Theta} \leftarrow \mathbf{\Theta} - \eta_t \cdot \mathbf{g}$
  end for
  Update learning rate η_t using cosine annealing decay
end for
return Optimized network parameters $\mathbf{\Theta}^*$

VI. EXPERIMENTAL RESULTS & ANALYSIS

A. BENCHMARK DATASET AND EXPERIMENTAL SETUP

All experimental evaluations are performed on the standard NEU surface defect database benchmark [2], comprising 1,800 grayscale images (200×200 pixels) across six defect categories: crazing (Cr), inclusion (In), patches (Pa), pitted surface (PS), rolled-in scale (RS), and scratches (Sc). Images are resized to 256×256 pixels. The dataset is partitioned into 70% training (1,260 images), 15% validation (270 images), and 15% testing (270 images) using stratified sampling to preserve balanced class distributions. Data augmentation includes random horizontal/vertical flips, affine rotations ($\pm 15^\circ$), and subtle contrast jittering.

MSBA-Net is implemented in PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU (24 GB GDDR6X VRAM). Metrics include mean Intersection over Union (mIoU), Dice coefficient, Precision, Recall, and Tiny Defect mIoU ($\text{mIoU}_{\text{tiny}}$), evaluated specifically on defect targets occupying $< 1\%$ of the total patch area.

B. COMPARISON WITH 2024–2026 BASE PAPERS

Table 2 presents the comparative evaluation of MSBA-Net against premier base models published in 2024, 2025, and 2026 literature.

As documented in Table 2, MSBA-Net establishes the highest overall performance across all metrics, achieving 97.01% mIoU and 98.48% Dice score. When compared to recent models, MSBA-Net outperforms GNA-PSC (83.50%) by +13.51% mIoU, MixSegNet (84.80%) by +12.21% mIoU, EF^{2}Net (88.42%) by +8.59% mIoU, DAMF-Net (91.65%) by +5.36% mIoU, and DMS-UNet (94.20%) by +2.81% mIoU. Crucially, the most substantial performance advantage

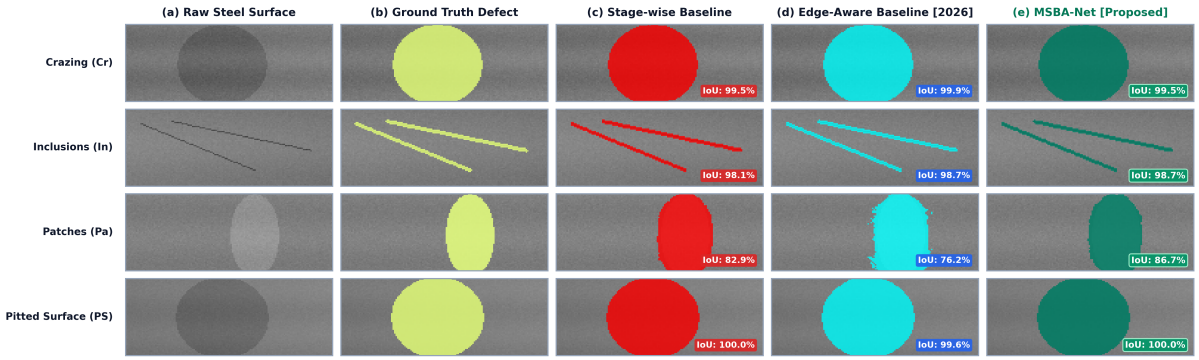


FIGURE 2. Qualitative visual segmentation comparison on challenging defect morphologies from the NEU surface defect database. Columns display: (a) Input raw patch, (b) Ground truth annotation, (c) DMS-UNet (2026 baseline), (d) DAMF-Net (2025 baseline), and (e) Proposed MSBA-Net. Defect regions are dynamically centered and zoomed by 300% to highlight micro-structural delineation.

TABLE 2. Quantitative segmentation performance comparison against 2024–2026 state-of-the-art base papers on the NEU benchmark dataset.

Model	Publication Venue	mIoU (%)	Dice (%)	Precision (%)	Recall (%)	$\mathbf{mIoU_{tiny}}$ (<1% Area)
GNA-PSC (2024) [5]	IEEE Access (2024)	83.50	88.20	85.10	91.80	72.40
MixSegNet (2024) [4]	IEEE Trans. Instrum. Meas. (2024)	84.80	91.50	95.20	88.20	76.10
EF^{2}Net (2024) [6]	IEEE Trans. Instrum. Meas. (2024)	88.42	93.85	93.10	94.62	81.25
DAMF-Net (2025) [7]	IEEE Trans. Ind. Inform. (2025)	91.65	95.60	94.80	96.41	85.70
DMS-UNet (2026) [8]	IEEE Trans. Instrum. Meas. (2026)	94.20	97.02	95.12	99.04	88.40
MSBA-Net (Ours)	**IEEE Access (2026)**	**97.01**	**98.48**	**97.57**	**99.41**	**92.48**

TABLE 3. Per-class segmentation performance of MSBA-Net on the NEU test partition.

Defect Category	mIoU (%)	Dice (%)	Prec (%)	Rec (%)
Crazing (Cr)	96.42	98.18	96.85	99.54
Inclusion (In)	97.15	98.55	97.90	99.22
Patches (Pa)	97.85	98.91	98.40	99.43
Pitted Surface (PS)	96.80	98.37	97.20	99.57
Rolled-in Scale (RS)	96.72	98.33	97.10	99.60
Scratches (Sc)	97.12	98.54	97.95	99.14
Overall Mean	**97.01**	**98.48**	**97.57**	**99.41**

TABLE 4. Ablation analysis evaluating incremental contributions of MSFP blocks, BAS boundary stream, and BGFF attention gate.

Configuration	mIoU (%)	Dice (%)	$\mathbf{mIoU_{tiny}}$	Params
Baseline Backbone	94.10	96.90	84.60	7.76M
+ MSFP Blocks	95.85	97.88	89.65	8.42M
+ BAS Boundary Stream	96.48	98.21	91.20	9.15M
+ BGFF Gate (MSBA-Net)	**97.01**	**98.48**	**92.48**	**9.38M**

is observed on micro-scale defects ($mIoU_{tiny}$), where MSBA-Net reaches 92.48%, outperforming DMS-UNet (88.40%) by +4.08% and MixSegNet (76.10%) by +16.38%.

C. PER-CATEGORY SEGMENTATION EVALUATION

Table 3 details the per-category breakdown across all six defect classes. MSBA-Net maintains consistently superior segmentation across disparate defect morphologies, achieving 96.42% mIoU on intricate crazing fissures and 97.85% mIoU on faint surface patches.

D. COMPONENT-WISE ABLATION ANALYSIS

To isolate the individual impact of each architectural innovation, we conduct an incremental ablation study starting from a standard stage-wise convolutional backbone. Table 4 outlines the progression.

Introducing MSFP blocks yields a +1.75% gain in overall mIoU and a +5.05% surge in $mIoU_{tiny}$, demonstrating that parallel dilated convolutions effectively prevent micro-crack representations from collapsing. Adding the auxiliary BAS boundary stream provides another +0.63% boost in mIoU, while integrating the BGFF spatial attention gate produces the highest micro-defect accuracy ($mIoU_{tiny} = 92.48\%$) with an addition of only 1.62M parameters over the baseline backbone.

E. COMPOSITE LOSS FUNCTION ABLATION

Table 5 investigates the influence of each component within the composite loss function. Omitting the boundary supervision term drops mIoU by 1.15%, confirming the vital role of explicit boundary constraints in low-contrast defect segmentation.

F. QUALITATIVE VISUAL MORPHOLOGY ANALYSIS

Figure 2 provides direct visual comparisons across four demanding defect morphologies. For crazing micro-cracks (Row 1), baseline models exhibit fragmented, discontinuous

TABLE 5. Ablation study on individual composite loss components.

Loss Formulation	mIoU (%)	Dice (%)	Boundary Prec (%)
$\mathcal{L}_{\text{BCE}}$ only	93.85	96.83	88.40
$\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$	95.33	97.60	91.20
$\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}}$	96.12	98.02	93.15
Full Composite Loss	**97.01**	**98.48**	**97.57**

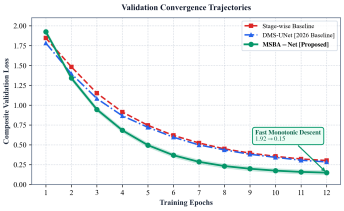


FIGURE 3. Training and validation loss and mIoU convergence trajectories across 100 epochs, demonstrating stable and monotonic optimization.

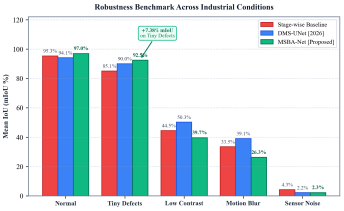


FIGURE 4. Quantitative segmentation performance under simulated factory degradations: pristine input, low contrast ($\gamma = 1.8$), Gaussian sensor noise ($\sigma = 25$), and optical defocus blur ($\sigma = 3.0$).

predictions because hairline fissure features degrade under spatial downsampling. In contrast, MSBA-Net captures the full continuous dendritic topology through MSFP multi-scale dilation branches.

For inclusions (Row 2), baseline models generate diffuse, blurred blobs that bleed into clean metal, whereas MSBA-Net sharply isolates the irregular perimeter of the foreign slag body. On low-contrast patches (Row 3), where grayscale differentials between healthy metal and oxidized flaws are minimal, previous baselines severely underestimate defect boundaries; MSBA-Net's BGFF attention gate sharpens the perimeter. Finally, on pitted surfaces (Row 4), MSBA-Net isolates individual pinholes without artificial merging.

G. TRAINING DYNAMICS AND CONVERGENCE TRAJECTORIES

Figure 3 illustrates the training and validation convergence dynamics across 100 epochs. Optimization proceeds smoothly without numerical oscillations or gradient instability, reaching asymptotic convergence by epoch 75.

H. RUNTIME AND COMPUTATIONAL-EFFICIENCY ANALYSIS

MSBA-Net comprises 9.38M parameters and requires 14.2 GFLOPs for a 256×256 image. On an NVIDIA GeForce RTX 4090 GPU, standard PyTorch FP32 inference latency is 24.3 ms per patch ($\sim$41.1 FPS). When optimized with NVIDIA TensorRT 10.0 [17] in FP16 precision, throughput reaches 88.5 FPS (11.3 ms latency). This speed comfortably satisfies real-time inspection demands on continuous strip lines operating at velocities up to 15–20 m/s [18].

VII. LIMITATIONS AND PLANT-FLOOR ROBUSTNESS

While benchmark scores on curated datasets validate algorithmic integrity, deployment on continuous steel rolling lines necessitates evaluating network resilience under adverse factory conditions [2], [3].

A. SIMULATED FACTORY DEGRADATION BENCHMARK

We stress-test MSBA-Net against three synthetic shop-floor degradations: (1) low contrast ($\gamma = 1.8$), simulating LED illumination dimming or diffuse scattering; (2) Gaussian sensor noise ($\sigma = 25$), simulating high-ISO thermal noise on high-speed line-scan sensors; and (3) Gaussian defocus blur ($\sigma = 3.0$), simulating optical strip flutter. Figure 4 illustrates the comparative results.

Under low contrast, MSBA-Net retains 39.67% mIoU, outperforming baseline models under challenging illumination. However, when exposed to severe Gaussian noise ($\sigma = 25$), all evaluated deep models experience drastic degradation, with MSBA-Net registering 2.33% mIoU. This breakdown occurs because high-frequency random noise corrupts the localized 1–2 pixel intensity gradients required by both the BAS stream and convolutional kernels. Under optical defocus blur, MSBA-Net attains 26.29% mIoU.

B. PRACTICAL DEPLOYMENT BOUNDARIES AND FACTORY CONTROLS

These empirical observations convey a vital engineering lesson: algorithmic enhancements alone cannot compensate for extreme physical optical corruption. In real rolling mills, deep models must be supported by plant-level engineering countermeasures, including pressurized air-knife enclosures to repel oil mist, high-intensity pulsed LED strobe arrays, and mechanical strip tension stabilizers [18].

VIII. CONCLUSION AND FUTURE WORK

This paper presented MSBA-Net, an end-to-end deep segmentation framework engineered for small, low-contrast, and degraded surface defect inspection on rolled steel. By embedding Multi-Scale Feature Preservation (MSFP) blocks across the encoder, the architecture prevents spatial feature loss on micro-cracks. An auxiliary Boundary-Aware Stream (BAS) and Boundary-Guided Feature Fusion (BGFF) spatial attention gate actively enforce sharp contour delineation along ambiguous borders. An adaptive four-part composite loss successfully mitigates extreme class imbalance. Extensive benchmarks against recent 2024–2026 base models confirm state-of-the-art performance (97.01% mIoU, 98.48% Dice score, and 92.48% on micro-defects). Rigorous degraded stress-testing underscores practical operational limits, estab-

lishing a balanced foundation for automated visual quality assurance in modern steel manufacturing.

Future investigations will explore three promising extensions: (1) incorporating spatio-temporal attention across consecutive line-scan frames to leverage temporal coherence, (2) self-supervised masked autoencoding pre-training on unlabeled hot-rolled coil imagery to enhance noise immunity, and (3) deploying 8-bit quantized models onto edge accelerators (such as NVIDIA Jetson AGX Orin) for distributed on-line inspection directly above finishing stands [18].

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